MACSE502 Programming for Data Science

YouTube Search Links by Syllabus Subtopic

73 clickable searches covering modules 1 to 5. Module 6 lists no topics in the syllabus.

What these links are. Each one opens YouTube's own search results for that subtopic. They are search pages, not specific videos: individual video links could not be checked when this file was made, so none are listed and nothing here is a recommendation of a particular video. A printed copy keeps the search words but loses the click, so the words are enough to type into YouTube.

Choosing a video. For Python, pandas and Scala, prefer a recent and longer tutorial, because library APIs change over the years. Check that the video uses the version you use in the lab (Scala 2.13, Spark 3.5). Module 5 is about opinion and trade-offs, so your lecture notes are the better source there.

Related subtopics that the syllabus lists separately are combined in one row, for example "Singleton" and "Companion Object". Module 5 lists 14 criteria; they are grouped into related searches because a single criterion such as "Purpose" gives poor results.

Module 1: Python Data Structures (8 h)

Subtopic	Open on YouTube (click the search words)
Condition and Branching Statements	python if elif else conditional statements tutorial
List, Tuple, Dictionary, Set	python list tuple dictionary set tutorial
Stack, Queue, Priority Queue	python stack queue priority queue heapq deque tutorial
String handling methods	python string methods tutorial
Exception Handling	python exception handling try except finally tutorial
Object-Oriented Concepts	python object oriented programming classes inheritance tutorial
File Handling	python file handling read write tutorial
APIs and Data Collection; Simple API and REST APIs	python requests REST API json tutorial
NLTK	nltk python tutorial tokenization stemming stop words
ChatterBot	python chatterbot chatbot tutorial

Module 2: Python Libraries (9 h)

Subtopic	Open on YouTube (click the search words)
Series	pandas series tutorial
DataFrame	pandas dataframe tutorial
Handling Missing Values	pandas handling missing values fillna dropna
Built-in functions; Data Operations	pandas describe groupby sort_values apply tutorial
Filtering Data in DataFrame; Data Extraction	pandas filter rows loc iloc query tutorial
Working with Text Data	pandas string methods str contains extract tutorial
Merging DataFrames	pandas merge join concat tutorial
Pandas (whole library)	pandas full course python data analysis
NumPy	numpy tutorial arrays python
SciPy	scipy tutorial python
Scikit-Learn	scikit-learn tutorial python machine learning
XGBoost	xgboost python tutorial
Keras	keras tutorial neural network python
TensorFlow	tensorflow tutorial beginners
PyTorch	pytorch tutorial beginners
Beautiful Soup	beautiful soup python web scraping tutorial
Scrapy	scrapy python web scraping tutorial
Matplotlib	matplotlib tutorial python
Seaborn	seaborn tutorial python
Bokeh	bokeh python interactive visualization tutorial
Plotly	plotly python tutorial
Folium	folium python map tutorial

Module 3: Scala Data Structures and OOP (12 h)

Subtopic	Open on YouTube (click the search words)
Whole module: Scala for beginners	scala full course for beginners
Expanded Function Format	scala function definition def tutorial
Variables, Numbers, Variable types	scala val var data types tutorial
Strings	scala strings string interpolation tutorial
Getting user input	scala readLine user input tutorial
Operators, Booleans	scala operators boolean tutorial
Arrays	scala array tutorial
Scala Array-REPL	scala REPL tutorial
Lists	scala list tutorial
Tuple	scala tuple tutorial
Sets, Hash set	scala set hashset tutorial
Maps	scala map tutorial
Functional Combinators	scala map filter reduce fold flatMap tutorial
Scala Object and Class	scala class object tutorial
Anonymous object	scala anonymous class object tutorial
Singleton, Companion Object	scala singleton object companion object tutorial
Case Classes	scala case class tutorial
Constructors	scala constructors primary auxiliary tutorial
Method Overloading	scala method overloading tutorial
This Keyword	scala this keyword tutorial
Inheritance	scala inheritance tutorial
Field Overriding, Final	scala override val final keyword tutorial
Abstract Class	scala abstract class tutorial
Trait, Trait Mixins	scala traits mixin tutorial
Access Modifiers	scala access modifiers private protected tutorial

Module 4: Scala Libraries and Spark Basics (8 h)

Subtopic	Open on YouTube (click the search words)
Breeze	scala breeze library tutorial
Saddle	scala saddle library series frame
Exception Handling (Scala)	scala exception handling try catch finally Try Option Either
Apache Spark Architecture	apache spark architecture driver executor cluster manager
Spark Big Data	big data introduction hadoop vs spark
Apache Spark Components	apache spark components spark core sql streaming mllib graphx

Module 5: Programming language selection criteria (6 h)

Subtopic	Open on YouTube (click the search words)
All criteria, overview	how to choose a programming language for a project criteria
Performance, Efficiency	programming language performance comparison python java scala
Security	choosing a programming language security considerations
Skill Set, Programmer experience, Support and community	choosing a programming language team skills community support
Targeted platform, Size of the Deployment, Purpose	choosing programming language target platform deployment requirements
Elasticity of a language	programming language flexibility scalability comparison
Time to production, Ease of Development and Maintenance	time to market programming language maintainability
Availability of an IDE	best IDE for python and scala comparison
Error Checking and Diagnosis	static vs dynamic typing compile time vs runtime errors
Applied: Python vs Scala for data work	python vs scala for data engineering and spark

Module 6: Contemporary Issues (2 h)

The syllabus lists no topics for this module, so there is nothing to search for.